BUILDLAB ENGINEERING

RC Engineering Toolkit

Starter Edition • Speed, gearing and build planning

Purpose

This starter toolkit helps hobbyists estimate theoretical RC vehicle speed, record build specifications, and plan controlled tests. It is an educational planning aid—not a substitute for manufacturer specifications, engineering validation, or safe operation.

Included

- Speed and gearing formulas
- Worked example with sample values
- Build planning worksheet fields
- Pre-test safety and validation checklist

BuildLab Engineering • Starter resource for feedback and iteration

1. Speed and gearing formulas

Total reduction ratio = motor revolutions per output-wheel revolution. For a single pinion/spur stage, ratio = spur gear tooth count ÷ pinion gear tooth count. Multiply ratios for additional reduction stages.

Wheel RPM = motor RPM ÷ total reduction ratio.

Wheel circumference = π × wheel diameter. If diameter is in millimetres, circumference is in millimetres.

Idealized speed (km/h) = wheel RPM × circumference (mm) × 60 ÷ 1,000,000.

Rough adjusted estimate = idealized speed × assumed efficiency fraction. For example, 85% becomes 0.85. This is only a rough illustration: a single efficiency percentage does not fully model motor load, tire slip, voltage sag, tire growth/deformation or aerodynamic drag.

2. Worked example

Assume motor speed = 6,000 RPM, total reduction ratio = 3.0:1, wheel diameter = 65 mm, and a rough efficiency assumption = 85%.

Quantity	Calculation	Result
Wheel RPM	6,000 ÷ 3.0	2,000 RPM
Circumference	π × 65 mm	204.2 mm
Idealized speed	2,000 × 204.2 × 60 ÷ 1,000,000	24.50 km/h
Rough adjusted estimate	24.50 × 0.85	20.83 km/h

The example produces about 20.83 km/h under the stated simplified assumptions. This is not a prediction of guaranteed measured speed; no-load motor RPM is often higher than loaded RPM.

3. Design notes and common mistakes

• Use the actual total reduction ratio, including every gearbox or belt stage.

• Confirm whether the motor RPM value is a no-load rating or a measured loaded value.

• Use the loaded voltage and realistic operating conditions when refining estimates.

• Do not exceed motor, electronic speed controller (ESC), battery, connector or wire ratings.

• Changing gearing changes both speed and torque at the wheels; a taller gear can increase current and heat under load.

• Treat efficiency as a rough placeholder until measured data is available.

4. Build planning worksheet

Fill in the companion CSV worksheet before a test. Record both expected values and measured outcomes; this makes it easier to learn from each iteration.

Planning field	Record
Vehicle / scale	_____
Motor and rated voltage	_____
Battery chemistry / cell count / capacity	_____
ESC current rating / motor current rating	_____
Pinion / spur tooth counts	_____
Other reduction stages	_____
Wheel diameter (mm)	_____
Estimated speed and assumptions	_____
Test conditions / safety area	_____
Observed speed / measurement method	_____
Motor / ESC temperature after test	_____
Battery voltage before and after	_____
Next change to test	_____

5. Pre-test checklist

- ■ Inspect gears, wheels, fasteners and drivetrain alignment.
- ■ Check battery condition, connector polarity, wire insulation and secure mounting.
- ■ Confirm voltage and current limits for the motor, ESC and battery.
- ■ Keep hands, loose clothing and tools away from moving parts; raise drive wheels for initial checks.
- ■ Test in a clear, controlled area away from people, roads and fragile property.
- ■ Stop if there is unusual noise, smoke, swelling, excessive heat or unstable handling.
- ■ Allow components to cool and record results before changing one variable at a time.

Disclaimer: This guide provides simplified educational calculations only. It does not certify a design or guarantee performance or safety. Follow component manuals and applicable local rules. Never charge or operate damaged batteries; use appropriate battery-safety procedures.

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